

Protected Attention–Koopman Takagi–Sugeno Fuzzy Models for Fault Detection and Margin-Certified Isolation in Closed-Loop Nonlinear Systems

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Abstract—Fault detection and isolation for closed-loop nonlinear processes must usually be designed from normal operating data alone, and any online monitor faces a structural dilemma: the normal-behavior reference must be corrected to prevent drift, yet every correction channel is a path through which faulty measurements can contaminate the reference and silently absorb the fault. This paper develops a protected monitoring architecture around a learned attention–Koopman Takagi–Sugeno (T–S) fuzzy model of the normal closed loop, in which a free-running reference branch is initialized only at delay-confirmed safe anchors and never reads post-anchor measurements. The stacked multi-step residual of the protected branch is shown to admit a structural decomposition into an anchor-error subspace, input-scaled model-mismatch and context-mismatch terms, stochastic innovations, and three fault-signature operators assembled online from Jacobian–vector products of the learned model. This decomposition yields the three main contributions: (i) a post-filter for the stacked residual that annihilates the anchor-error subspace, whitens across time, and augments the omnibus statistic with matched signature branches; (ii) an online dynamic threshold generator that propagates input-dependent mismatch envelopes directionally through the membership-scheduled dynamics and calibrates the observable excess score by split-conformal quantiles, giving a distribution-free pointwise false-alarm guarantee together with a class-conditional minimum detectable fault; and (iii) a structured isolation layer with explicit actuator, sensor, and process signatures, channel masking formalized as structural annihilation, and a decision rule whose correctness is certified by pairwise principal-angle margins, with explicit rejection when the margins fail. A complete evaluation protocol on a closed-loop continuous stirred-tank reactor benchmark is specified.

Index Terms—Fault detection, fault isolation, Takagi–Sugeno fuzzy systems, Koopman operator, dynamic threshold, conformal calibration, closed-loop systems.

I. INTRODUCTION

SAFETY-critical industrial processes operate in closed loop, traverse multiple nonlinear operating regimes, and must be monitored for actuator, sensor, and process faults whose signatures are not known in advance. In most applications the data available for monitor design are records of normal operation only: faults are rare, expensive to reproduce, and almost never labeled. A practical diagnosis scheme therefore has to learn what normal closed-loop behavior looks like, detect departures from it with a controlled false-alarm rate, and attribute a detected departure to a structural fault class—all without ever having seen a fault.

Three methodological traditions address parts of this problem. Model-based residual generation—observer-based and parity-space design, residual evaluation, and threshold computation—provides the most complete theory [1]–[5], but presupposes an explicit dynamic model with a priori uncertainty bounds. Multivariate statistical monitoring builds data-driven normal-behavior models, yet its classical instruments are static projections that ignore closed-loop dynamics [6], [7]. Between the two, learned dynamic models have emerged: Koopman-operator liftings represent nonlinear dynamics by (bi)linear evolution in an observable space [8]–[11], T–S fuzzy structures preserve local-linear models under nonnegative memberships that sum to one [12], and attention encoders summarize input–output histories into latent states [13]. Recent work applies learned Koopman models to fault diagnosis with parity or observer residuals trained on fault-free data [14]–[16].

The first unresolved difficulty is reference contamination. Every online monitor corrects its normal-behavior reference with incoming measurements—through an observer gain, a parity stack over the latest window, or an encoder that re-reads the measurement history. After a fault occurs, the same channel drags the reference toward the faulty plant, the residual shrinks, and a persistent or slowly developing fault is absorbed rather than detected. Observer-corrected and measurement-driven parity residuals of learned Koopman models [15], [16] inherit this channel by construction. The opposite extreme—a permanently free-running simulation—avoids contamination but accumulates unbounded prediction error [17]. What is missing is an architecture that separates the two needs in time: corrections restricted to anchors confirmed as fault-free after a delay, no measurement influence on the reference after the anchor, and a finite-horizon error theory that quantifies for how long the free-running reference stays trustworthy.

The second difficulty concerns the evaluation of stacked multi-step residuals. When the reference is a free-running rollout, the natural residual is the whole window of multi-step prediction errors, and its fault-free value is driven by propagated anchor uncertainty, input-dependent model mismatch, and serially correlated noise. Classical dynamic thresholds propagate a priori uncertainty bounds through known LTI or LPV residual dynamics [18]–[21]; set-based thresholds propagate zonotopes or intervals through given analytic T–S or LPV models [22]–[26]. None of these constructions applies

when the model itself is learned, the mismatch envelope must be estimated from normal data as a function of the measured input, and the residual is a stacked window whose rows share the same disturbance realizations. Conversely, conformal methods deliver distribution-free false-alarm control for arbitrary scores [27]–[30], but calibrate the raw score directly and discard the model structure that makes a threshold input-aware. A principled division of labor between a model-based deterministic threshold generator and a distribution-free probabilistic layer is missing.

The third difficulty is structured isolation. Structured and directional residuals, incidence matrices, and unknown-input observer banks solve isolation when fault entry maps are known [2], [31]–[33]; data-driven constructions exist for linear systems [14], [34]; learned-model diagnosis either isolates within a single fault category [15] or relies on empirical channel masking without guarantees [35]. Quantitative isolability has been studied through distinguishability measures and residual-direction limits [36], [37], and principal angles between actuator and sensor signature subspaces have recently been used for nonlinear systems with learned observer components [38], though without a process class, an only-normal-data protocol, or rejection semantics. What is absent is a class-level theory—actuator versus sensor versus process—for residuals of learned closed-loop models: explicit signature operators, conditions under which the classes are distinguishable, a margin that can be evaluated online, and an honest rejection output when the margin fails.

This paper addresses the three difficulties jointly, for closed-loop nonlinear systems and under a strict only-normal-data protocol. The key observation is that all three reduce to properties of one object: the stacked residual of a protected free-running rollout, together with a small family of structured operators—anchor-error propagation, innovation propagation, and fault signatures—that are computable online from Jacobian–vector products of the learned model. The contributions are:

- 1) *Protected residual generation.* A dual-branch architecture in which a data branch encodes the strict past of the measured closed loop while a protected branch rolls the learned attention–Koopman T–S model freely from delay-confirmed safe anchors; a state machine guarantees that no post-anchor measurement can reach the long-term reference, and an exact propagation identity expresses the stacked residual through segment-averaged fuzzy Jacobians.
- 2) *Post-filter design for the stacked residual.* A first-order structural decomposition of the stacked fault-free residual is derived, with an explicitly quantified bilinear remainder. On its basis a three-stage post-filter is designed: annihilation of the rank- m_z anchor-error subspace (a parity-space projection transplanted from measured-output stacks [39] to protected rollouts), cross-time whitening, and a branch bank that retains a full-rank omnibus statistic while adding matched subspace statistics [40] for the structured fault classes; the information cost of the projection is characterized exactly.
- 3) *Input-scheduled dynamic threshold with a distribution-*

free guarantee. An online threshold generator propagates normal-data-estimated, input-dependent mismatch envelopes directionally through the membership-scheduled Jacobians, preserving cross-time correlation over the stacked window; the observable excess over this deterministic part is calibrated by split-conformal quantiles on held-out normal blocks, yielding a pointwise false-alarm guarantee that survives envelope misspecification, together with a class-conditional minimum detectable fault.

- 4) *Margin-certified structured isolation.* Explicit signature operators are assembled online: a transition-weighted block-Toeplitz actuator signature, a two-path banded sensor signature obtained from encoder Jacobian–vector products, and a process class characterized as state-channel injection outside the actuator range; channel masking is formalized as structural annihilation of the encoder-mediated sensor path; a decision theorem certifies correct classification whenever pairwise principal-angle margins between whitened signature subspaces exceed the nuisance level, and prescribes rejection otherwise.

Throughout, the closed-loop compensation limit is stated explicitly: both branches are conditioned on the same recorded commands, so controller-mediated fault compensation cancels from the residual to first order and genuinely reduces detectability; the proposed thresholds make this loss observable rather than hidden.

The remainder of the paper is organized as follows. Section II reviews controlled Koopman liftings and T–S fuzzy structure, states the system and information model, and formulates the two design objectives. Section III develops the protected architecture, the structural decomposition and post-filter, the dynamic threshold generator, the isolation layer, and the learning procedure. Section IV specifies the evaluation protocol on a closed-loop continuous stirred-tank reactor benchmark. Section V concludes. Long proofs are collected in the Appendix.

Notation: $\mathbb{R}^m$ denotes m -dimensional Euclidean space and $\|\cdot\|$ the Euclidean norm or the induced spectral norm. Dimensions carry the prefix m (e.g., m_z), k indexes samples, and s denotes an anchor instant. Boldface lowercase letters are stacked vectors and boldface capitals stacked operators, displayed with explicit brackets at first use. Calligraphic capitals denote mappings, and blackboard-bold capitals denote sets, spaces, and subspaces. The fault family is reserved for f and its capital forms; noise and disturbance terms use ε with roman superscripts. For a matrix M , M^T , $M^\dagger$, $\sigma_{\min}(M)$, and $\text{Range}(M)$ are its transpose, pseudoinverse, smallest singular value, and column space. I_m is the $m \times m$ identity, and Π_V the orthogonal projector onto a subspace V .

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Controlled Koopman Lifting

Consider a discrete-time controlled nonlinear system $x_{k+1} = \mathcal{X}(x_k, u_k, \xi_k)$ with state $x_k \in \mathbb{R}^{m_x}$, control input $u_k \in \mathbb{R}^{m_u}$, and measured exogenous condition $\xi_k \in \mathbb{R}^{m_\varepsilon}$. Let

$\mathbb{H}^\infty$ be a Hilbert space of observables $\psi : \mathbb{R}^{m_x} \rightarrow \mathbb{R}$. The controlled Koopman operator family $\mathcal{K}_{(u,\xi)} : \mathbb{H}^\infty \rightarrow \mathbb{H}^\infty$ acts by composition with the frozen-input transition,

$$(\mathcal{K}_{(u,\xi)}\psi)(x) = \psi(\mathcal{X}(x, u, \xi)), \quad (1)$$

so that the nonlinear state evolution becomes linear evolution of observables for each frozen input pair [8]. Finite-dimensional computation replaces (1) by a controlled linear predictor on a learned lifting $\Psi : \mathbb{R}^{m_x} \rightarrow \mathbb{R}^{m_z}$,

$$z_{k+1} = Az_k + B_u u_k + B_\xi \xi_k, \quad \hat{y}_k = C_z z_k, \quad (2)$$

with $z_k \approx \Psi(x_k) \in \mathbb{R}^{m_z}$, matrices $A \in \mathbb{R}^{m_z \times m_z}$, $B_u \in \mathbb{R}^{m_z \times m_u}$, $B_\xi \in \mathbb{R}^{m_z \times m_\xi}$, and an output map $C_z \in \mathbb{R}^{m_y \times m_z}$; (2) is a genuine approximation, obtained in practice by extended dynamic mode decomposition or by jointly learning Ψ and the matrices [9]–[11]. A single linear predictor, however, cannot represent strongly regime-dependent dynamics, which motivates the fuzzy combination reviewed next.

B. T-S Fuzzy Combination of Local Koopman Models

A T-S fuzzy model [12] interpolates N_{rule} local models through memberships defined on a premise variable $\rho \in \mathbb{R}^{m_\rho}$. Throughout we require the memberships $\omega_i(\rho)$ to satisfy the standard conditions

$$\omega_i(\rho) \geq 0, \quad \sum_{i=1}^{N_{\text{rule}}} \omega_i(\rho) = 1. \quad (3)$$

In this paper the memberships are *learned*: a premise network $\mathcal{P}$ maps the available latent information to ρ , rule scores are quadratic in ρ with learnable prototypes and bounded-spectrum metrics, and ω_i is their softmax, so (3) holds by construction and all maps are continuously differentiable. The T-S Koopman transition used throughout is

$$\mathcal{Z}_\Theta(z, u, \xi; \chi) = \sum_{i=1}^{N_{\text{rule}}} \omega_i(\rho) (A_i z + B_i u + o_i), \quad (4)$$

where $\rho = \mathcal{P}(z, u, \xi)$, the pair (A_i, B_i, o_i) is the i th local Koopman model, and $\chi \in \mathbb{R}^{m_\chi}$ is an attention context defined in Section III. The complete Jacobians of (4), including the membership-variation term, are given in Section III-B; they are the objects propagated by the threshold generator.

C. System, Faults, and Information Model

Assumption 1 (Plant and fault channels). The monitored closed-loop plant evolves as

$$\begin{aligned} u_k^{\text{act}} &= u_k + f_k^a, \\ x_{k+1} &= \mathcal{X}^n(x_k, u_k^{\text{act}}, \xi_k) + f_k^p + \varepsilon_k^x, \\ y_k &= Cx_k + f_k^s + \varepsilon_k^y, \end{aligned} \quad (5)$$

where $\mathcal{X}^n$ is the unknown fault-free transition mapping, u_k is the command recorded by the controller, $y_k \in \mathbb{R}^{m_y}$ the measured output, $f_k^a \in \mathbb{R}^{m_u}$, $f_k^p \in \mathbb{R}^{m_x}$, and $f_k^s \in \mathbb{R}^{m_y}$ are actuator, process, and sensor faults, and $\varepsilon_k^x, \varepsilon_k^y$ are process disturbance and measurement noise. In fault-free operation $f_k^a = f_k^p = 0$ and $f_k^s = 0$.

Assumption 2 (Information boundary). Online, the monitor trusts the recorded command u_k , the measurement y_k , and the exogenous condition ξ_k ; the applied input u_k^{act} , the state x_k , and all fault signals are unavailable. The exogenous condition is assumed unaffected by faults. Model training, structure and hyperparameter selection, and all threshold calibration use normal data only, partitioned into disjoint training, validation, and calibration blocks plus a frozen test segment; fault data enter only the final frozen test.

Assumption 3 (Normal operating domain). The learned model, once frozen, is continuously differentiable on a compact normal latent domain, and the fault-free forcing terms defined in Section III-B admit envelopes estimated on normal validation data. All guarantees are stated conditionally on these envelopes and are flagged as such.

The closed-loop setting deserves emphasis. The controller maps measurements to commands, so a sensor fault also perturbs future commands. Because the monitor conditions *both* of its branches on the same recorded command sequence, the controller's response is explained by the model and cancels from the residual to first order; this removes the need to know the control law but also means that fault energy compensated by the controller is genuinely invisible. This information-theoretic loss is a property of the setting, not of a particular algorithm, and is kept explicit throughout.

D. Problem Formulation

Let $\mathcal{D}_n = \{(u_k, y_k, \xi_k)\}$ denote the normal dataset of Assumption 2, and let $\alpha \in (0, 1)$ be a prescribed false-alarm level. The paper solves the following two problems.

Problem 1 (*protected detection with guaranteed false-alarm rate*). Design, using $\mathcal{D}_n$ only, a residual generator $k \mapsto e_k$ whose normal-behavior reference is provably not corrupted by post-anchor measurements, a post-filtered statistic T_k , and an online threshold γ_k^{dyn} varying with the measured input, the operating regime, and the reference age, such that in fault-free operation

$$\Pr \{T_k > \gamma_k^{\text{dyn}}\} \leq \alpha, \quad (6)$$

and such that, for each structured fault class c , an explicit energy level guarantees $T_k > \gamma_k^{\text{dyn}}$.

Problem 2 (*margin-certified structured isolation*). After an alarm, design statistics $\{T_c\}$ over the classes $c \in \{\mathbf{a}\} \cup \{\mathbf{s}-j\}_{j=1}^{m_y} \cup \{\mathbf{p}\}$ and an online-computable margin κ_k such that, whenever a single-class fault of sufficient energy is active and the margin condition holds, the decision rule returns the true class,

$$\arg \max_c T_c = c^* \quad \text{whenever} \quad \|\mathbf{v}_{c^*}\| \geq \frac{2\varepsilon_n}{1 - \kappa_k}, \quad (7)$$

where $\mathbf{v}_{c^*}$ is the whitened fault signature and ε_n the calibrated nuisance level, and returns an explicit rejection verdict whenever the margin condition fails.

III. PROPOSED METHOD

A. Protected Dual-Branch Residual Generation

The monitor maintains two strictly separated information paths built on one learned model $\{\mathcal{E}, \mathcal{P}, \mathcal{Z}_\Theta, C_z\}$.

1) *Data branch*: At every instant the causal attention encoder $\mathcal{E}$ reads the strict-past history

$$d_k^- = [u_{k-\ell_h}^\top \ y_{k-\ell_h}^\top \ \cdots \ u_{k-1}^\top \ y_{k-1}^\top \ \xi_{k-\ell_h}^\top \ \cdots \ \xi_k^\top]^\top \quad (8)$$

and returns the data-branch latent state and attention context,

$$(z_k^E, \chi_k^E) = \mathcal{E}(d_k^-) \in \mathbb{R}^{m_z} \times \mathbb{R}^{m_\chi}. \quad (9)$$

The current measurement y_k is deliberately excluded from d_k^- : a sensor abnormality at instant k cannot enter the latent state before $k+1$, the current output residual retains direct sensitivity to it, and the isolation layer of Section III-D obtains a clean temporal ordering of fault propagation.

2) *Protected branch*: Let s be a *safe anchor*, an instant whose data have been confirmed fault-free by the delayed-confirmation rule below. The protected branch initializes $\hat{z}_{s|s} = z_s^E$ and rolls the model freely,

$$\hat{z}_{t+1|s} = \mathcal{Z}_\Theta(\hat{z}_{t|s}, u_t, \xi_t; \chi_{t|s}^n), \quad \hat{y}_{t|s} = C_z \hat{z}_{t|s}, \quad (10)$$

where the protected context $\chi_{t|s}^n$ is produced by $\mathcal{E}$ from a protected history buffer that contains the *predicted* outputs $\hat{y}_{\cdot|s}$ together with the recorded u, ξ ; no measurement taken after s ever enters (10). The recorded command u_t is shared by both branches, which is what removes the unknown control law from the analysis and, at the same time, causes the closed-loop compensation loss discussed in Section II.

3) *Anchor state machine*: A candidate anchor at instant s is *confirmed* only at instant $s+N_c$, and only if no warning or alarm occurred in $(s, s+N_c]$; confirmed anchors advance the long-term reference, and re-anchoring is performed by a fixed, alarm-independent schedule interleaved with this confirmation rule, so that the calibration data can replay the identical policy. Let N_a denote the maximal detector reaction delay to any fault that will ever be flagged.

Proposition 1 (Contamination blocking). *If $N_c \geq N_a$, then every measurement that influences the long-term protected reference was generated at least N_c samples before any instant at which the detector reacts to a fault; consequently no measurement produced by a detectable fault can reach the protected reference before the alarm. Faults slower than the confirmation horizon and below the warning sensitivity remain outside this guarantee and are listed as an explicit blind class.*

The proof is immediate from the construction and is omitted. The constant N_c is set from the evidence-propagation length—the history depth ℓ_h plus the lifted order m_z plus a small safety margin—so that its choice requires no fault data.

4) *Unified residual and exact propagation*: For any start instant s (a training start, the short-horizon start $k-\ell_s$, or the long-term anchor s_k^n), define the residual, the one-step innovation, and the context mismatch

$$\begin{aligned} e_{k|s} &= z_k^E - \hat{z}_{k|s}, \\ \nu_k &= z_{k+1}^E - \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E), \\ \delta_{k|s}^\chi &= \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_k^E) - \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^n), \end{aligned} \quad (11)$$

together with the output residual $r_k^y = y_k - C_z z_k^E$. All four quantities are online computable, and the same rollout (10) is

used in training and online, which removes the distribution mismatch between teacher-forced training and free-running monitoring.

Proposition 2 (Exact stacked propagation). *Let $\mathcal{P}$ be continuously differentiable, and define the segment-averaged Jacobian and transition products*

$$\begin{aligned} A_{k|s} &= \int_0^1 J_z \mathcal{Z}_\Theta(\hat{z}_{k|s} + q e_{k|s}, u_k, \xi_k; \chi_{k|s}^n) dq, \\ \Phi(k, j) &= A_{k-1|s} \cdots A_{j|s}, \quad \Phi(k, k) = I_{m_z}. \end{aligned} \quad (12)$$

Then, for every $l \geq 1$,

$$e_{s+l|s} = \sum_{j=0}^{l-1} \Phi(s+l, s+j+1) (\nu_{s+j} + \delta_{s+j|s}^\chi), \quad (13)$$

an exact identity that holds for arbitrary fault action on the data branch, because no fault quantity was differentiated in its derivation.

Stacking the window rows of (13) gives the central object of the paper. With $e_{s,N} = [e_{s+1|s}^\top \cdots e_{s+N|s}^\top]^\top \in \mathbb{R}^{Nm_z}$ and ν, δ stacked likewise,

$$e_{s,N} = G_\nu (\nu + \delta), \quad [G_\nu]_{l,j} = \begin{cases} \Phi(s+l, s+j), & j \leq l, \\ 0, & j > l, \end{cases} \quad (14)$$

where $[G_\nu]_{l,j} \in \mathbb{R}^{m_z \times m_z}$ denotes the (l, j) block. Since the diagonal blocks equal I_{m_z} , the operator G_ν is unit block lower triangular and invertible: the stacked residual and the innovation sequence carry the same information, and the choice between them is a choice of coordinates. The complete fuzzy Jacobians needed in (12), including the membership-variation term $\Delta_\rho J_{\rho,z}$ that dominates near rule boundaries, are assembled analytically from the softmax gradient identity and are stated with their dispersion bounds in the Appendix (Proposition 7); their neglect systematically underestimates the propagation gain, which is why all operators in this paper are built from the complete Jacobians.

B. Structural Decomposition and Post-Filter Design

This subsection derives the structure of the fault-free stacked residual and designs the evaluation filter on it. Throughout, s is the window anchor and N the window length; all operators are assembled online along the protected trajectory from Jacobian–vector products of the frozen model.

1) Context cancellation and the combined forcing:

Lemma 1 (Context cancellation). *For every k and start s , $\nu_k + \delta_{k|s}^\chi = z_{k+1}^E - \mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^n)$: the dependence on the data-branch context χ_k^E cancels algebraically, at every fault amplitude.*

The proof is one line from (11). The stacked identity (14) therefore depends on the data branch only through the latent sequence z^E and on the protected branch through the normal context; in particular, fault energy appearing in δ^χ alone is an artifact of the additive split and carries no isolation information (Section III-D).

2) *First-order structure of the fault-free window*: Write $z_k^E = z_k^* + \varepsilon_k^e$, where z^* is the noise-free encoding of the underlying normal trajectory and ε_k^e collects the measurement- and encoder-induced per-step error, including the context-noise term $J_\chi \mathcal{Z}_\Theta \cdot (\partial_\chi / \partial Y) \varepsilon^y$ of the data branch. Expanding the combined forcing of Lemma 1 to first order and telescoping the ε^e chain through G_ν yields the following.

Proposition 3 (Structural decomposition with quantified remainder). *Let L_H be a Lipschitz constant of $z \mapsto J_z \mathcal{Z}_\Theta$ on the normal domain. In fault-free operation,*

$$e_{s,N} = \varepsilon_+^e - G_0 \varepsilon_s^e + G_\nu (\varepsilon^m + \varepsilon^v + d^x) + r^{\text{rem}}, \quad (15)$$

where $\varepsilon_+^e = [(\varepsilon_{s+1}^e)^\top \dots (\varepsilon_{s+N}^e)^\top]^\top$ carries the per-step encoding errors, $G_0 = [\Phi(s+1, s)^\top \dots \Phi(s+N, s)^\top]^\top \in \mathbb{R}^{Nm_z \times m_z}$ propagates the anchor-instant error through the whole window, $\varepsilon^m, \varepsilon^v, d^x$ are the stacked model-mismatch, process-noise, and context-drift forcings, and the remainder obeys $\|r^{\text{rem}}\| = O(L_H (\|\varepsilon^e\|^2 + \|\varepsilon^e\| \|e\|))$. Under active faults, the window gains the signature terms

$$e_{s,N} += F_a f^a + \sum_{j=1}^{m_y} F_s^{(j)} f_j^s + G_\nu w^p, \quad (16)$$

with $F_a = G_\nu \text{blkdiag}(B_s^u, \dots, B_{s+N-1}^u)$ the transition-weighted input-response operator, $F_s^{(j)}$ the banded operator assembled from the encoder Jacobian–vector products with respect to the channel- j measurement-history slots (including, for pre-anchor onsets, the protected-context-buffer slots), and w^p an unknown state-channel forcing representing process faults.

Three structural facts in (15)–(16) drive everything that follows. First, the anchor-error term $-G_0 \varepsilon_s^e$ occupies a subspace of dimension at most m_z , is perfectly correlated across all window rows, and grows with the transition products—it is precisely the term whose scalar upper bound dominates radius-recursion thresholds at long reference ages. Second, actuator and process faults are *propagated* forcings, filtered by G_ν , whereas sensor faults enter through the encoder image of the history as *banded* perturbations of width ℓ_h that do not accumulate along the window; this asymmetry, a direct consequence of the protected branch never reading post-anchor measurements, is the structural basis of isolation. Third, the remainder is *bilinear*: its constant grows with the sharpness of the rule metrics (through L_H), so the decomposition—and every statement built on it—carries the explicit validity condition that the metric sharpness times the normal error envelope be small; the monitor evaluates a membership-dispersion gate online and degrades to the guard statistic below whenever the gate fails. In the numerical verification of Section IV, the identity (14) holds to 10^{-15} , the anchor term is first-order exact with the predicted bilinear scaling, and the banded sensor structure is reproduced with machine-zero out-of-band response.

3) *Stage 1: anchor annihilation*:

Proposition 4 (Anchor annihilator). $\text{rank } G_0 = \text{rank } A_s$, where A_s is the pointwise Jacobian at the anchor (the

segment average degenerates to it because $e_{s|s} = 0$). Let $Q \in \mathbb{R}^{Nm_z \times (Nm_z - r_0)}$, $r_0 = \text{rank } G_0$, be an orthonormal basis of $\text{Range}(G_0)^\perp$. Then $Q^\top e$ is free of the anchor-error term up to the bilinear remainder, and the information removed by $Q^\top$ is characterized exactly: a fault pattern is invisible to all annihilated statistics if and only if its window signature lies in $\text{Range}(G_0)$; in particular, by the factorization $\Phi(s+l, s) = \Phi(s+l, s+1)A_s$, the response of an actuator fault acting only at the first post-anchor step satisfies $\text{Range}(F_a^{(1)}) \subseteq \text{Range}(G_0)$ and is annihilated exactly, and the retained energy fraction of early-step signatures decreases with the accumulated transition gain.

The annihilation is therefore a deliberate and quantified trade: it removes the dominant, unbounded-in-age nuisance term at the cost of first-post-anchor-step fault sensitivity. The design does not hide this cost but covers it: the branch bank below retains an *un-annihilated guard statistic*—the raw stacked norm evaluated against a conservative radius-recursion threshold—whose only purpose is to see what $Q^\top$ discards; its false alarms are priced by the same conformal layer. In implementation the projection is energy-thresholded: only directions of $\text{Range}(G_0)$ whose gain exceeds the noise floor are annihilated, which also disposes of the near-singular- A_s ambiguity.

4) *Stage 2: cross-time whitening*: The covariance Σ_e of the stacked fault-free residual is estimated once per fuzzy region on the estimation block with Ledoit–Wolf shrinkage, a ridge, and an eigenvalue floor [41]; each window then uses $\Sigma_Q = Q^\top \Sigma_e Q$ and its inverse square root $\widetilde{W} = \Sigma_Q^{-1/2}$. Serial correlation of the encoding errors (each measurement enters up to ℓ_h encoder windows) is absorbed by Σ_e itself; no independence across rows is assumed anywhere, since the false-alarm guarantee is carried by the conformal layer. Estimating an $Nm_z \times Nm_z$ covariance requires the sample sizes quantified in the protocol; regions that fall short shrink harder toward the pooled estimate, and the affected cells are reported.

5) *Stage 3: branch bank*: Detection uses a bank of statistics on each window, all computed unconditionally at every step so that their calibration on normal data requires no alarm-conditioning:

$$\begin{aligned} T_{0,k} &= \|\widetilde{W} Q^\top e\| && \text{(omnibus, annihilated),} \\ T_{g,k} &= \|e\| && \text{(guard, un-annihilated),} \\ T_{c,k} &= \|\Pi_{S_c} \widetilde{W} Q^\top e\| && \text{(matched, } c \in \mathcal{C}), \end{aligned} \quad (17)$$

where the matched classes $\mathcal{C}$ use the *persistent* low-dimensional signature subspaces—actuator bias $S_a = \text{Range}(\widetilde{W} Q^\top F_a (\mathbf{1}_N \otimes I_{m_u}))$ of dimension at most m_u , per-channel sensor bias, and persistent process forcing as in Section III-D. Restricting the matched branches to low-dimensional persistent parameterizations is what makes them worthwhile: under Gaussian whitened noise each branch is the generalized-likelihood-ratio detector for a mean shift in its subspace [40], and the energy required for detection scales with the square root of the branch dimension, giving the persistent branches an order- $\sqrt{Nm_z/m_u}$ advantage over the omnibus statistic, whereas full-class subspaces of dimension

growing with N would forfeit the gain. Every branch is evaluated against its own deterministic normalization and the shared conformal layer of Section III-C; the max over branches is calibrated once, so adding branches trades a small uniform quantile inflation for large directional gains, with no silent blind spot: the guard branch sees the annihilated directions, and the omnibus branch sees everything else.

C. Input-Scheduled Dynamic Threshold With a Distribution-Free Guarantee

The evaluation of the post-filtered statistics separates two responsibilities that the literature usually conflates. A deterministic *threshold generator* propagates normal-data-estimated envelopes of the structured nuisance terms through the actual window operators, producing an online threshold that follows the measured input, the operating regime, and the reference age; a *conformal layer* then calibrates the observable excess over this deterministic part on held-out normal blocks. The division of labor is stated once and precisely: the false-alarm guarantee of Theorem 1 rests on exchangeability alone and survives misspecification of every envelope; the envelopes are accountable only for detection power and for the detectability statement of Corollary 1, where their validity at fault-time inputs enters as an explicit assumption. The deterministic part is thus a normalization surface with physical meaning—it is what makes the threshold input-scheduled—while the distribution-free layer owns the guarantee.

1) *Input-dependent envelopes*: On the estimation block, the mismatch magnitude $\|\varepsilon_j^m\|$ is regressed against the features $(1, \|u_j\|, \|\Delta u_j\|, \|\Delta \xi_j\|)$ by a nonnegative quantile fit within each fuzzy region c , giving the online envelope

$$\bar{\mu}_j = \eta_0(c_j) + \eta_u(c_j)\|u_j\| + \eta_{\Delta u}(c_j)\|\Delta u_j\| + \eta_{\xi}(c_j)\|\Delta \xi_j\|, \quad (18)$$

so the measured instantaneous input enters the threshold with a known slope. The context-drift magnitude is enveloped by an age curve $\bar{\delta}(h)$ indexed by the anchor age h , estimated on the same block. Both are empirical quantile surfaces on the normal distribution and are labeled as such; their realized exceedance rate is monitored online as a self-diagnostic.

2) *Directional propagation over the stacked window*: Let $C_j \in \mathbb{R}^{(N-1)m_z \times m_z}$ denote the j th block column of $Q^T \mathbf{G}_\nu$, i.e., the propagation of the step- j forcing into the entire annihilated window, and let $\tilde{W}$ be the stacked whitening of Section III-B. The deterministic threshold is

$$\gamma_{k,N}^{\det} = \sum_{j=1}^N (\bar{\mu}_j + \bar{\delta}(h_j)) \|\tilde{W} C_j\|, \quad (19)$$

computed by a few power iterations per block column and updated incrementally as the window slides.

Proposition 5 (Containment for the first-order surrogate). *If $\|\varepsilon_j^m + d_j^x\| \leq \bar{\mu}_j + \bar{\delta}(h_j)$ for all window steps j , then the deterministic part of the first-order surrogate of Proposition 3 satisfies $\|\tilde{W} Q^T \mathbf{G}_\nu(\varepsilon^m + d^x)\| \leq \gamma_{k,N}^{\det}$.*

The proof is the triangle inequality over block columns and is exact for the surrogate; the bilinear remainder of

Proposition 3 is not covered by (19) and is absorbed by the conformal layer below. Two structural properties distinguish (19) from scalar-radius recursions: each source j enters as one vector through the true operators, so cancellations induced by the annihilator and by the whitening geometry are captured within each source; and the anchor term, which dominates scalar recursions at long reference ages, is absent because Q removed it.

Remark 1 (Honest comparison with scalar recursions). The bound (19) is not pointwise tighter than a scalar-radius recursion: summing block-column norms is worst-case across sources, and in fast-mixing regimes with near-isotropic whitened geometry a root-sum-of-squares recursion can win by up to $\sqrt{N}$. The advantages of (19) are the removal of the anchor radius and the directional evaluation through $\tilde{W}$; in the numerical study of Section IV the combined effect roughly halves the threshold at window length $N = 8$, and the ablation A3 attributes the gain between the two mechanisms.

3) *Conformal layer and the false-alarm guarantee*: For each branch b of the bank (structure $\times$ scale), define the observable excess score $\tau_{b,k} = \max\{0, T_{b,k} - \gamma_{b,k}^{\det}\}/\sigma_b$, with σ_b a per-branch scale from the estimation block, and let $\tau_k = \max_b \tau_{b,k}$. On the calibration block—disjoint from the estimation block, with the re-anchoring policy replayed—the scores are formed into non-overlapping blocks of length exceeding the residual mixing time, each block is reduced to its maximum, and the scores are stratified into Mondrian cells by fuzzy region and reference-age group. The online threshold is

$$\gamma_{b,k}^{\text{dyn}} = \gamma_{b,k}^{\det} + \sigma_b q_{1-\alpha}(c_k, h_k), \quad (20)$$

where $q_{1-\alpha}(c, h)$ is the conformal rank quantile of the cell; cells with fewer than $\lceil 1/\alpha \rceil - 1$ block scores return $+\infty$ and are merged by proximity, with all merges reported.

Theorem 1 (Pointwise false-alarm guarantee). *Suppose (i) the four normal blocks of Section III-E are disjoint and frozen, so that the score function—including $\tilde{W}$, the envelopes, and the branch scales—is fixed before calibration; (ii) within each Mondrian cell, the calibration block-maximum scores and the online block-maximum score are exchangeable in fault-free operation; and (iii) the online re-anchoring policy equals the calibration replay policy and is alarm-independent. Then, in fault-free operation,*

$$\Pr \{\tau_k > q_{1-\alpha}(c_k, h_k)\} \leq \alpha, \quad (21)$$

i.e., the alarm rule $\exists b : T_{b,k} > \gamma_{b,k}^{\text{dyn}}$ has pointwise false-alarm level α . The guarantee does not depend on the correctness of the envelopes (18), whose misspecification degrades power and conditional balance but not (21).

The proof, in the Appendix, is the split-conformal rank argument [27] applied to the block-maximum scores, with the observable-excess construction replacing the unobservable noise residual; time dependence is handled by the block reduction, and deviations from exact exchangeability translate into a coverage gap bounded by the total-variation drift between blocks [30]. Because marginal coverage does not

imply conditional coverage [42], false alarms may cluster in regimes where the envelope underfits; the protocol therefore reports per-cell empirical coverage, and an out-of-coverage monitor on the operating descriptor (a second conformal test on normal data) withholds all certified verdicts outside the covered domain.

4) Class-conditional detectability:

Corollary 1 (Minimum detectable persistent fault). *Let class c have whitened annihilated signature $\tilde{\mathbf{F}}_c = \tilde{\mathbf{W}}\mathbf{Q}^\top \mathbf{F}_c$, and let the fault be persistent over the window, $\mathbf{f} = \mathbf{1}_N \otimes f_0$. Assume the envelopes (18) hold along the fault-perturbed trajectory (Assumption A5', stated in the Appendix) and the calibration event of Theorem 1 holds. If*

$$\sigma_{\min}(\tilde{\mathbf{F}}_c(\mathbf{1}_N \otimes I)) \|\mathbf{f}_0\| \geq 2(\gamma_{c,k}^{\det} + \sigma_c q_{1-\alpha}(c_k, h_k)), \quad (22)$$

then $T_{c,k} \geq \gamma_{c,k}^{\text{dyn}}$, and at least one window in any sliding sequence covering the fault raises the branch above its threshold.

The restriction to the persistent-fault subspace $\mathbf{1}_N \otimes I$ is essential: over the full column space, $\sigma_{\min}$ is driven to zero by transient patterns concentrated in the final window step (and vanishes identically when $Nm_u > (N-1)m_z$), whereas such patterns are recovered by the sliding window itself, becoming early-step faults of later windows. When the alarm logic additionally requires m -of- n persistence, (22) guarantees the exceedances, not the latched alarm, and the delay accounting in Section IV reflects this.

Remark 2 (Adaptive blindness is observable, not absent). A fault that drives the controller raises $\|u\|, \|\Delta u\|$ and hence inflates $\gamma^{\det}$ with the known slopes of (18); the conformal quantile can change only by switching Mondrian cells, a precomputed bounded step. The total signature energy that can be masked this way is bounded by the input-saturation budget multiplied by the envelope slopes plus the largest inter-cell quantile gap—an explicit stealthy-fault set that the paper reports rather than hides. The input-driven threshold contribution is logged at every alarm and every suppressed alarm, and the out-of-coverage monitor bounds how far the excursion can travel while still claiming certified operation.

D. Margin-Certified Structured Isolation

Isolation runs after an alarm, on a fresh local window anchored at the last confirmed safe anchor, and uses the signature operators of Proposition 3. Its logic rests on the exact cancellation of Lemma 1—sensor faults reach the stacked residual only through the banded encoder paths, the current-output residual r^y , and the shared recorded-command path—together with one structural zero and one subspace-margin theorem.

1) Masking as structural annihilation:

Proposition 6 (Mask-conditioned exoneration). *Let the mask-conditioned encoder zero channel j at the raw input before any cross-channel mixing, use per-channel precomputed normalization, and let the premise input and ξ contain no function of y_j . If the entire pipeline—anchor encoding, both branches, and both context buffers—is re-run under the mask Λ^{-j} , then*

the Jacobian of every encoder-mediated path from y_j to the masked stacked residual e^{-j} is identically zero: the sensor- j signature is removed structurally, at every fault amplitude, not attenuated. The only surviving paths of f_j^s into e^{-j} are second-order terms through the recorded commands.

The exoneration test compares the masked omnibus statistic against its mask-specific calibrated threshold: a drop below threshold under Λ^{-j} , while the unmasked statistic alarms, supports the verdict Sensor- j . The evidence is deliberately one-sided: because the command path carries a bias quadratic in the fault amplitude, a large sensor fault can keep e^{-j} above threshold, so a failed exoneration must never exclude the sensor class; it routes the case to the mixed-fault test below. The amplitude ceiling at which the command-path bias reaches the threshold is calibrated from normal-data curvature probes and reported with every verdict.

2) *Structured classes and the margin theorem:* The certified classes are the proper-subspace classes: actuator, with $\mathbb{S}_a = \text{Range}(\tilde{\mathbf{W}}\mathbf{Q}^\top \mathbf{F}_a)$; each sensor channel, with $\mathbb{S}_{s-j} = \text{Range}(\tilde{\mathbf{W}}\mathbf{Q}^\top \mathbf{F}_s^{(j)})$ assembled from both banded paths; and the *persistent process* class $\mathbb{S}_p = \text{Range}(\tilde{\mathbf{W}}\mathbf{Q}^\top \mathbf{G}_\nu(\mathbf{1}_N \otimes I_{m_z}))$, which encodes the declared prior that a process fault acts as a slowly varying state-channel forcing over the window. No claim is made that arbitrary process faults lie in $\mathbb{S}_p$; fast or erratic process action falls to the uncertified energy test below. For each class the statistic is the dimension-normalized projection energy $T_c = \|\Pi_{\mathbb{S}_c} \tilde{\mathbf{r}}\|$ with a per-class calibrated noise quantile ε_c , where $\tilde{\mathbf{r}} = \tilde{\mathbf{W}}\mathbf{Q}^\top \mathbf{e}$ is the whitened annihilated residual of the isolation window.

Theorem 2 (Margin-certified isolation). *Let a single-class fault of class c^* act on the isolation window, $\tilde{\mathbf{r}} = \mathbf{v} + \mathbf{n}$ with $\mathbf{v} \in \mathbb{S}_{c^*}$ and $\|\mathbf{n}\| \leq \varepsilon_n$, and let $\kappa = \max_{c \neq c'} \cos \theta_{\min}(\mathbb{S}_c, \mathbb{S}_{c'})$ be the largest pairwise principal-angle cosine among the certified class subspaces, computed online from their bases. If*

$$\|\mathbf{v}\| > \frac{2\varepsilon_n}{1-\kappa}, \quad (23)$$

then $\arg \max_c T_c = c^$. If (23) cannot be certified—because κ approaches one, because the two largest statistics differ by less than $2\varepsilon_n$, or because the noise level exceeds its calibrated envelope—the isolation layer returns the explicit verdict NONUNIQUE instead of a class.*

The proof (Appendix) is the standard subspace-perturbation argument: $T_{c^*} \geq \|\mathbf{v}\| - \varepsilon_n$, while every competing projection is bounded by $\kappa\|\mathbf{v}\| + \varepsilon_n$. The margin κ and the per-decision gap are computed and reported at decision time, so every verdict carries its own certificate.

3) *Decision procedure:* The full layer proceeds as follows. (0) Out-of-coverage check: outside the certified operating domain only the detection alarm is maintained. (1) Temporal-profile statistic: the growth of row energies across nested horizons—banded signatures plateau after ℓ_h rows, propagated signatures accumulate—is scored against its normal calibration; this is a calibrated discriminator, not a subspace fact, and only shapes priorities. (2) Sensor stage: per-channel matched statistics, the exoneration test of Proposition 6, the direct-

feedthrough consistency of r_j^y , and the channel-redundancy floor $\text{PI}_j \geq \text{PI}_{\min}$. (3) Actuator/process stage: T_a with channel-level margins, T_p on the persistent-process subspace, and the residual energy outside all structured subspaces against its own quantile, whose exceedance yields the uncertified verdict PROCESS-SIDE. (4) Mixed faults by *masked deflation*: if the sensor- j statistic is high and the masked pipeline still alarms with an actuator or process match on e^{-j} , the verdict MIXED(sensor- j , c) is returned; the deflation test runs before margin rejection, and NONUNIQUE is preferred whenever it is inconclusive. (5) Any failed premise—coverage, redundancy, margin, amplitude ceiling—produces the corresponding rejection verdict rather than a forced classification. The identifiability boundary is inherited unchanged: a process fault whose window action lies in the actuator response range is indistinguishable from an equivalent input deviation, and the reported quantity is an equivalent input deviation, never a physical actuator-fault confirmation.

Remark 3 (Validity envelope of the sensor stage). The matched sensor statistic relies on first-order encoder signatures evaluated in the normal context; its validity window is bounded by the fault amplitude (linearization), by the bilinear leakage that grows with the accumulated residual, and by the loop time constant (command-path bias); the exoneration test is amplitude-independent on all encoder-mediated paths and is therefore the robust arm of the stage. For faults with onset before the isolation anchor, the protected context buffer holds measured samples for the first ℓ_h window rows; the corresponding buffer path is part of the sensor signature, and the anchor-cleansing effect of the annihilator holds to first order in the context sensitivity rather than exactly.

E. Learning and Calibration Protocol

All model components are trained jointly on the normal training block. The history (8) is tokenized per instant, embedded linearly, and processed by a causal multi-head attention encoder with smooth activations, so that every Jacobian used above exists and is continuous; the premise network is a small bottleneck MLP; the rule metrics use a bounded-spectrum parameterization; the local matrices A_i carry a soft spectral cap as a numerical safeguard; and the output map C_z is linear, which the isolation layer exploits. Training minimizes a free-running multi-horizon loss: anchors s and horizons $N \sim \text{Unif}\{1, \dots, N_{\max}\}$ are sampled, the same rollout (10) used online produces $\hat{z}_{s+j|s}$ and $\hat{y}_{s+j|s}$ without reading intermediate measurements, and weighted latent and output prediction errors are accumulated, together with a decoding-consistency term and a variance regularizer that jointly exclude latent collapse, a membership-balance term that keeps all rules in use without forbidding sharp partitions, and a finite-horizon product penalty on the complete Jacobian norms that discourages explosive rollouts. Channel masking is applied during training with a fixed distribution (all-pass, single-channel, and independent multi-channel masking), which trains the shared mask-conditioned encoder used for isolation without degrading the unmasked task.

Algorithm 1 Protected monitoring at instant k

- 1: encode $d_k^- \rightarrow (z_k^E, \chi_k^E)$; advance short- and long-term rollouts (10)
 - 2: update ν_{k-1} , $\delta_{k-1|s}^x$, r_k^y , and the window stack $e_{s,N}$
 - 3: assemble/update G_ν , G_0 , and the signature blocks by JVPs; refresh Q
 - 4: compute branch statistics T_0, T_a, T_{s-j} and the threshold γ_k^{dyn}
 - 5: update warning/alarm state machine; confirm or advance anchors
 - 6: **if** alarm **then**
 - 7: run the isolation layer of Section III-D; report class or rejection with margins
 - 8: **end if**
-

Calibration discipline is stricter than usual and is essential for the guarantee of Section III-C: the normal data are split into *four* disjoint frozen blocks—(i) a training block for all learnable parameters; (ii) an estimation block on which the mismatch envelopes, the context-drift curve, the stacked covariances, and the branch scales are estimated; (iii) a calibration block on which only the conformal quantiles are computed; and (iv) a held-out normal test segment that joins the frozen fault test set. Fitted preprocessing is learned on the training block only; isolation gaps of length $\ell_h + N_{\max}$ separate consecutive blocks; and the online re-anchoring policy is replayed identically on the calibration block so that reference ages are equally distributed. No fault data influence any of these steps.

F. Online Operation and Complexity

Per instant, the dominant costs are one encoder pass, two rollout steps, the analytic Jacobian assembly (a small number of vector-Jacobian products of $\mathcal{P}$), and incremental updates of the block columns of G_ν ; the annihilator refresh is a thin SVD of the $Nm_z \times m_z$ matrix G_0 . Signature assembly by JVPs runs only after alarms. For the benchmark dimensions of Section IV the whole loop is dominated by the encoder pass and is real-time capable by a wide margin; a detailed operation count is given in the supplementary protocol.

IV. EXPERIMENTAL PROTOCOL

The method targets closed-loop process data under the only-normal-data protocol of Assumption 2. This section fixes the complete evaluation protocol. In accordance with that protocol, all numerical entries are produced by a single frozen evaluation run executed after every design quantity of Sections III–III-F has been frozen; the result placeholders below are intentionally left open in this manuscript version and no number reported here was obtained from fault data.

A. Benchmark and Data Split

The primary benchmark is a closed-loop continuous stirred-tank reactor (CSTR) simulation dataset with commands $u = (C_i, T_i, T_{ci})$ and measured outputs $y = (C, T, T_c, Q_c)$, one

long normal record, and eight faulty records with a common fault onset; faults 1–2 are process faults, 3–5 actuator faults, and 6–8 sensor faults, so the benchmark covers exactly the three structured classes of Section III-D. The normal record is partitioned 55/15/20/10% into the four blocks of Section III-E with isolation gaps; the faulty records join the frozen test set with per-sample labels beginning at the true onset.

B. Numerical Design Verification

Independently of the benchmark, every structural claim of Section III was verified numerically on randomized synthetic T–S Koopman models ($m_z = 6$, $m_u = 2$, four rules, window $N = 8$, double precision), with the following outcomes. The stacked identity (14) holds to a relative error of 4.5×10^{-15} under arbitrary forcing, and $Q^T G_0$ vanishes to machine precision. The anchor term of (15) is first-order exact in the clean regime (error ratio 100.0 across a decade of perturbation scales) and exhibits exactly the predicted bilinear cross-term scaling in the noisy regime (ratio 10.1); annihilation removes 99.7% of an injected anchor-perturbation energy under realistic forcing. The actuator signature is first-order accurate with the predicted $O(\|e\|)$ relative floor, the sensor response is banded with machine-zero out-of-band entries, and the mask-conditioned encoder has an exactly zero Jacobian column for the masked channel. The block-column threshold (19) contains 3000 Monte-Carlo worst-case draws (max-to-bound ratio 0.42) and undercuts the scalar-radius recursion by a factor of about two at $N = 8$; the margin condition of Theorem 2 produced 400/400 correct decisions above the certified energy level. These checks validate the algebra and the design mechanisms; they are not process-data results.

C. Baselines and Ablations

Baselines span the three traditions of Section I: static multivariate monitoring (PCA with T^2 /SPE, a reconstruction autoencoder), a one-step dynamic predictor with a constant calibrated threshold, and a measurement-driven Koopman parity design representative of [15], [16]. Ablations remove one design element at a time: (A1) no anchor annihilation; (A2) no matched branches (omnibus only); (A3) scalar-radius threshold in place of the block-column generator; (A4) conformal calibration of the raw score without the deterministic part; (A5) envelope without input features; (A6) isolation without masking exoneration; (A7) always-decide isolation without the margin-based rejection. Each ablation isolates one claim of Sections III-B–III-D.

D. Metrics

Detection is scored by false-alarm rate on fault-free segments (target α), missed-detection rate, detection delay from onset, and empirical in-control run length; threshold quality by the empirical coverage of γ_k^{dyn} on held-out normal data against its nominal level, reported per operating-regime cell to expose conditional-coverage clustering; isolation by the confusion structure over {actuator, sensor- j , process} with rejection reported as its own outcome (a rejected case is

TABLE I
DETECTION RESULTS ON THE FROZEN CSTR TEST SET (TO BE COMPLETED BY THE FROZEN EVALUATION RUN; NO ENTRY MAY BE FILLED FROM ANY RUN THAT INFLUENCED DESIGN).

Method	FAR	MDR	Delay	ARL ₀
PCA T^2 /SPE	–	–	–	–
Reconstruction AE	–	–	–	–
One-step predictor	–	–	–	–
Koopman parity [16]	–	–	–	–
Proposed (full)	–	–	–	–
Ablations A1–A5	–	–	–	–

TABLE II
ISOLATION RESULTS ON THE FROZEN CSTR TEST SET (TO BE COMPLETED BY THE FROZEN EVALUATION RUN; REJECTION IS REPORTED AS AN EXPLICIT OUTCOME).

Method	Correct	Wrong	Rejected	Margin κ
Proposed	–	–	–	–
A6 (no masking)	–	–	–	–
A7 (no rejection)	–	–	–	–

neither correct nor a misclassification, and the rejection rate is reported alongside), plus the online margin values at decision time; and the closed-loop compensation loss by the fraction of injected fault energy explained by the recorded commands. Placeholder summary tables (Tables I–II) fix the reporting format in advance.

V. CONCLUSION

This paper turned the informal requirement “learn normal behavior and monitor against it” into a protected architecture with three certified layers. The protected dual-branch design resolves the correction-versus-contamination dilemma in time, and the exact propagation identity reduces the monitoring problem to the structure of one stacked residual. On that structure, the anchor-annihilating post-filter with a guard branch converts the dominant nuisance of free-running references into a quantified, covered trade; the input-scheduled threshold generator with conformal excess calibration separates physical normalization from the distribution-free false-alarm guarantee and yields an online class-conditional minimum detectable fault; and the structured isolation layer replaces heuristic attribution with signature operators, a structural masking zero, and margin-certified decisions that refuse to guess when the geometry does not support a verdict.

The limitations are stated as design facts rather than caveats: fault energy compensated by the controller through the shared recorded commands is invisible to any method with this information set; the structural decomposition carries a bilinear validity gate that tightens near sharp rule boundaries and is monitored online; a stealthy fault set bounded by the envelope slopes and the inter-cell quantile gaps can inflate its own threshold within an explicit budget; and marginal false-alarm control does not imply conditional control, which the protocol therefore measures. Completion of the frozen benchmark evaluation of Section IV, extension of the guarantee layer to nonstationary normal behavior via weighted

exchangeability, and joint design of the re-anchoring schedule with the threshold generator are the immediate next steps.

APPENDIX

A. Complete Fuzzy Jacobians and Dispersion Bounds

Proposition 7 (Complete T-S Koopman Jacobians). *Let the memberships be the softmax of scores $\varsigma_i(\rho) = -\frac{1}{2}(\rho - p_i)^\top M_i(\rho - p_i) + o_i^\omega$ with $M_i = M_i^\top \succ 0$, and let $\mathcal{P}$ be continuously differentiable. With $\varsigma_i^\nabla = -M_i(\rho - p_i)$, $\bar{\varsigma}^\nabla = \sum_i \omega_i \varsigma_i^\nabla$, $\bar{v} = \sum_i \omega_i v_i$, and $v_i = A_i z + B_i u + o_i$,*

$$\begin{aligned} \nabla_\rho \omega_i &= \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla), \\ J_z \mathcal{Z}_\Theta &= A_\omega + \Delta_\rho J_{\rho,z}, \quad J_u \mathcal{Z}_\Theta = B_\omega + \Delta_\rho J_{\rho,u}, \\ \Delta_\rho &= \sum_i \omega_i (v_i - \bar{v}) (\varsigma_i^\nabla - \bar{\varsigma}^\nabla)^\top, \end{aligned} \quad (24)$$

with $A_\omega = \sum_i \omega_i A_i$ and $B_\omega = \sum_i \omega_i B_i$, and the dispersion bound

$$\|\Delta_\rho J_{\rho,z}\| \leq \sqrt{\lambda_{\max}(\Sigma_v)} \sqrt{\lambda_{\max}(J_{\rho,z}^\top \Sigma_\varsigma J_{\rho,z})}, \quad (25)$$

where $\Sigma_v = \sum_i \omega_i (v_i - \bar{v})(v_i - \bar{v})^\top$ and $\Sigma_\varsigma = \sum_i \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla)(\varsigma_i^\nabla - \bar{\varsigma}^\nabla)^\top$.

Proof. The softmax gradient follows from logarithmic differentiation. Differentiating $\mathcal{Z}_\Theta = \sum_i \omega_i v_i$ in z gives A_ω plus $\sum_i v_i (\nabla_\rho \omega_i)^\top J_{\rho,z}$; inserting the gradient identity and using $\sum_i \omega_i (\varsigma_i^\nabla - \bar{\varsigma}^\nabla) = 0$ centers v_i to $v_i - \bar{v}$, giving Δ_ρ . Writing $\Delta_\rho = V_w S_w^\top$ with $V_w = [\sqrt{\omega_i}(v_i - \bar{v})]_i$ and $S_w = [\sqrt{\omega_i}(\varsigma_i^\nabla - \bar{\varsigma}^\nabla)]_i$ and applying the submultiplicative bound $\|V_w S_w^\top J_{\rho,z}\| \leq \|V_w\| \|S_w^\top J_{\rho,z}\|$ yields (25). All expressions were verified against central differences to 10^{-10} . $\square$

The second term of (24) vanishes as the memberships approach one-hot and can dominate near rule boundaries; estimating the propagation gain by $\|A_\omega\|$ alone therefore systematically underestimates it. The same structure bounds the context sensitivity $\|J_\chi \mathcal{Z}_\Theta\|$, which enters the drift envelope $\bar{\delta}(h)$ and the buffer path of the sensor signature.

B. Proof of Proposition 2

By the fundamental theorem of calculus along the segment from $\hat{z}_{k|s}$ to z_k^E with the protected context frozen, $\mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^E) - \mathcal{Z}_\Theta(\hat{z}_{k|s}, u_k, \xi_k; \chi_{k|s}^n) = A_{k|s} e_{k|s}$ with $A_{k|s}$ as in (12). Adding and subtracting $\mathcal{Z}_\Theta(z_k^E, u_k, \xi_k; \chi_{k|s}^E)$ and using (11) gives the one-step recursion $e_{k+1|s} = A_{k|s} e_{k|s} + \nu_k + \delta_{k|s}^\chi$; no fault signal is differentiated because the identity manipulates definitions only. Unrolling from $e_{s|s} = 0$ yields (13), and stacking rows gives (14). $\blacksquare$

C. Proof Sketch of Proposition 3

Insert $z^E = z^* + \varepsilon^e$ into the combined forcing of Lemma 1 and expand to first order at z^* : the forcing splits into the ideal-trajectory mismatch and noise $\varepsilon_k^m + \varepsilon_k^x$, the context-drift term d_k^x , the encoder-noise difference $\varepsilon_{k+1}^e - J_z \mathcal{Z}_\Theta(z_k^*) \varepsilon_k^e$, and a Taylor remainder quadratic in ε^e . Propagating through (14) and telescoping $\sum_j \Phi(l, j+1)(\varepsilon_{j+1}^e - A_j \varepsilon_j^e) = \varepsilon_l^e - \Phi(l, s) \varepsilon_s^e + \rho_l$ leaves the telescoping defect ρ_l , which collects the products

$\Phi(l, j+1)(A_{j|s} - J_z \mathcal{Z}_\Theta(z_j^*)) \varepsilon_j^e$; since the segment-averaged and pointwise Jacobians differ by at most $L_H(\|e_{j|s}\| + \|\varepsilon_j^e\|)$, the defect and the Taylor remainder together obey the stated bilinear bound. The signature terms (16) follow by the same first-order expansion of the faulty data branch: actuator faults perturb the applied input at each step and propagate through G_ν ; sensor faults perturb the encoder inputs, whose effect telescopes to the direct banded image (plus the buffer path when the fault predates the anchor); process faults perturb the state channel and propagate through G_ν with unknown per-step injection. The numerical scaling study of Section IV-B confirms both the clean first-order rates and the bilinear floor. $\blacksquare$

D. Assumption A5' and Proofs for Section III-C

Assumption A5' (envelope validity at fault time). Along the fault-perturbed trajectory up to the decision instant, the fault-free forcing components still satisfy $\|\varepsilon_j^m + d_j^x\| \leq \bar{\mu}_j + \bar{\delta}(h_j)$ evaluated at the realized inputs and regimes, and the calibration event of Theorem 1 holds. This is an assumption about model validity off the normal input distribution; it is reported, not proved, and its online surrogate is the envelope-exceedance monitor.

Proof of Theorem 1. By (i) the score map $e \mapsto \tau$ is a fixed measurable function determined before the calibration block is touched; by (iii) the online anchor-age process equals the calibration replay in distribution. Consider the Mondrian cell of (c_k, h_k) and the block containing k . By (ii) the calibration block maxima $\tau^{(1)}, \dots, \tau^{(n_c)}$ and the online block maximum τ^* are exchangeable, so the rank of τ^* among them is uniform and $\Pr\{\tau^* > \tau^{((n_c+1)(1-\alpha))}\} \leq \alpha$ [27]. Since the pointwise score is dominated by its block maximum, the pointwise exceedance event is contained in the block-level event, giving (21). The envelopes enter the score only through the fixed map and therefore affect power, not coverage; deviations from exact block exchangeability add the total-variation drift bound of [30] to α . $\blacksquare$

Proof of Corollary 1. On the calibration event, the fault-free content of the branch statistic is at most $\gamma_{c,k}^{\det} + \sigma_{c,q_{1-\alpha}}$. By the reverse triangle inequality, $T_{c,k} \geq \|\Pi_{\mathbb{S}_c} \tilde{W} Q^\top F_c f\| - (\gamma_{c,k}^{\det} + \sigma_{c,q_{1-\alpha}})$, and for a persistent fault the projected signature norm is bounded below by $\sigma_{\min}(\tilde{F}_c(\mathbf{1}_N \otimes I)) \|f_0\|$ because the persistent signature lies in $\mathbb{S}_c$ by construction. Condition (22) then forces $T_{c,k} \geq \gamma_{c,k}^{\text{dyn}}$. $\blacksquare$

E. Proof of Theorem 2

Since $v \in \mathbb{S}_{c^*}$, $T_{c^*} = \|\Pi_{\mathbb{S}_{c^*}}(v + n)\| \geq \|v\| - \varepsilon_n$. For $c \neq c^*$, the largest principal-angle cosine bounds the cross-projection, $\|\Pi_{\mathbb{S}_c} v\| \leq \cos \theta_{\min}(\mathbb{S}_{c^*}, \mathbb{S}_c) \|v\| \leq \kappa \|v\|$, hence $T_c \leq \kappa \|v\| + \varepsilon_n$. Subtracting, $T_{c^*} - T_c \geq (1 - \kappa) \|v\| - 2\varepsilon_n > 0$ under (23) with strict inequality, so the argmax is unique and correct. The rejection rule is the contrapositive: when (23) cannot be certified from the online κ , the calibrated ε_n , and the observed gap, no class verdict is issued. $\blacksquare$

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